FAHIM AHMAD

Mechanical Design Engineer | Engineering Project Manager | Robotics & Automation | Design for Manufacturing

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Professional Summary

Mechanical Design Engineer and Engineering Project Manager with 3+ years delivering robotic systems, from parametric SolidWorks design held to **±0.1 mm** to **field-deployed EOD ROVs serving on UN peacekeeping missions**. Led **17 engineers** across **6 concurrent product lines**. CSWP plus five SOLIDWORKS Professional certifications.

Professional Experience

- **RB Labs (Robot Bulls)** Geneva, Switzerland (Consultancy)
  Mechanical Design Engineer, Humanoid Robotics *Aug 2026 – Present*
  - Design **parametric 3D CAD models of mechanical components and assemblies** for humanoid robot heads, balancing industrial design aesthetics against structural, functional, and manufacturability requirements
  - Develop **compact electronics packaging and housings** that integrate sensors and control electronics within the head assembly envelope
  - Conduct **R&D on materials and manufacturing processes** to optimise head assembly layouts
  - Review and incorporate **design feedback from engineering teams across Geneva and the USA** within a fully remote workflow, accelerating design iteration cycles

- **Cybernetics Hi-Tech Solutions (Pvt.) Ltd.** Dhaka, Bangladesh
  Engineering Project Manager, Robotics & Automation *Dec 2025 – Present*

  Portfolio: Smart Energy Meter | EOD ROV | AGV | UAV | Industrial 3D Printer | Automation Systems
  - Directed **6 concurrent hardware product lines**, managing end-to-end execution from R&D to field delivery with full accountability for milestones, deliverables, and stakeholder reporting across defence, energy, and industrial automation clients
  - Reduced delivery lead time by **12%** by restructuring procurement workflows, vendor coordination, factory visits, and logistics across local and international supply chains
  - Achieved **15% cost reduction** through BOM rationalization, vendor negotiations, finance process control, and dispute resolution across billing cycles, MoUs, and payment terms
  - Managed stakeholder relationships spanning defence procurement authorities, energy clients, and industrial automation partners, driving revenue conversion through technical demos, structured proposals, and post-delivery engagement
  - Established operational tracking via **MS Project** dashboards, improving cross-team visibility and enabling proactive risk mitigation across high-complexity mechatronics programmes

- **Walton Digi-Tech Industries Limited** Dhaka, Bangladesh
  First Senior Assistant Director, Mechanical Designer *Sep 2025 – Dec 2025*
  - Translated **product strategy into structured workstreams** with milestones, requirements, and cross-department deliverables aligned to **7 product lines**
  - Led **design reviews** and ensured requirement alignment across R&D, production, quality, and senior management, reducing review cycle time by **2.5%**
  - Delivered **engineering documentation, release packages, and change-control files** supporting **3 product families** for mass manufacturing readiness
  - Collaborated with **sales and marketing teams** to support go-to-market planning and product launch execution
  - Ensured **cross-functional alignment across Sales, Marketing, Supply Chain, and Engineering** to maintain product readiness and stakeholder visibility

- **Cybernetics Hi-Tech Solutions (Pvt.) Ltd.** Dhaka, Bangladesh
  Mechanical Team Lead, Robotics Projects *Jul 2024 – Sep 2025*

  Project: EOD ROV, Man-transportable Explosive Ordnance Disposal Vehicle

  Role: Mechanical Team Lead & PM | Team: 8 Engineers | Budget Authority: 4 Crore BDT
  - Secured government project approval by leading the bid team that defined **120+ technical specifications and 30+ key deliverables** with defence procurement authorities
  - Designed and validated actuation systems using **Harmonic and Planetary servo motors with CANBus protocols**, meeting torque and precision control requirements confirmed through kinematic analysis
  - Ensured compliance with **MIL-STD-810H** (environmental), **MIL-STD-461G** (EMI/EMC), and **MIL-STD-1472H** (human engineering) throughout design, validation, and risk assessment

- Cut development cycle time by **18%** through **FMEA-driven design validation**, ensuring on-milestone field deployment
- Reduced system cost by **16%** through BOM rationalization and vendor selection without compromising performance requirements

Project: Cyberfleet AGV System, RMG Factory Deployment
Role: Project Manager | Team: 6 Engineers | Live production at The Urmi Group

- Achieved a **13% reduction in material handling time** by redesigning sensor-optimised navigation routes and workflow maps for a 3-robot fleet in live production
- Designed **navigation logic and operational protocols** for AGV deployment in dynamic factory environments, with all milestones delivered on schedule
- Built the full BOM, project budget, and **KPI tracking framework**; conducted risk assessment and devised emergency operation protocols enabling safe autonomous deployment

Project: UAV Multi-Payload Platform (Surveillance | Cargo 10 to 100 kg | Combat-Support)
Role: Project Manager | Clients: Bangladesh Army, Border Guard Bangladesh and 3+ additional defence clients

- Reduced project costs by **10%** through BOM optimisation and resource allocation across three payload classes
- Coordinated development of **autonomy-capable UAV platforms**, translating specification feedback from **5+ clients** into validated engineering changes through iterative design reviews
- Maintained delivery schedule across concurrent payload development tracks through structured milestone tracking and cross-team dependency management

- **Spectrum Engineering Consortium (Pvt.) Ltd.** — Dhaka, Bangladesh
  R&D Engineer, Mechanical — *Apr 2023 – Jun 2024*

Project: EOD ROV "Jontro Soinik 1.0", Bangladesh's First Domestically Built and Exported Defence Robot
Role: Industrial Design and R&D Engineer | Clients: DGDP / Bangladesh Army Bomb Disposal Unit / UN Peacekeeping Mission

- **Delivered 3 field-deployed EOD ROVs** to DGDP, accepted against military-grade criteria and deployed on **UN Peacekeeping Missions in Mali and the Democratic Republic of Congo**
- System formally **transferred to the Peruvian Armed Forces on 05 May 2024**, Bangladesh's first export of a domestically built defence robot; facilitated **operational and maintenance training** for UN delegates and Peruvian Armed Forces personnel
- Achieved **±0.1 mm manufacturing tolerance** across **30+ mechanical components and prototypes** by building parametric SolidWorks models and GD&T-annotated engineering drawings for structural and actuation assemblies, optimising mass distribution, packaging, and thermal management
- Validated chassis and 6-DOF manipulator reliability under operational and shock loads through **ANSYS structural simulation** consistent with MIL-STD-810H, using workspace envelopes and torque requirements derived from kinematic and dynamic analysis
- Reduced late-stage design changes by **15%** by applying **FMEA** across **10+ formal design reviews** to verify design intent and production readiness
- Reduced assembly error rate by **23%** (26 to 20 defects per 100 assemblies) by writing **structured build instructions and in-process inspection protocols**, and resolved **30+ R&D blockers** with electronics, controls, and fabrication teams

Education

- **Ahsanullah University of Science and Technology (AUST)** — Dhaka, Bangladesh
  B.Sc. in Industrial and Production Engineering — CGPA: 3.51 / 4.00 — *January 2019 – May 2023*

Thesis: Mechanical properties of $ZrSiO_4$/PLA ceramic components fabricated by FDM; published in IMEOM 2022 proceedings

Skills

Mechanical Design & CAD: Parametric CAD Modeling, Assembly Design, Engineering Drawings, SolidWorks (CSWP), Sheet Metal, Weldments, Mold Tools, Fusion 360, SpaceClaim, Autodesk Design Modeler, Mesh-to-Parametric Conversion

Manufacturing Design: Design for Manufacturing (DFM/DFA), GD&T, Dimensional Tolerancing, Manufacturing Handoff, Electronics Packaging, Additive Manufacturing (FDM)

CAD Validation & QA: Design Reviews, Design Intent Verification, FMEA, In-Process Inspection Protocols, Production Readiness Checks

Simulation & Analysis: ANSYS Workbench, SolidWorks Simulation, FEA, Structural Analysis, Kinematic and Dynamic Analysis

Design Automation: SOLIDWORKS API, Python

Robotics & Systems: Humanoid Robot Hardware, 6-DOF Kinematics, Control Systems, Sensor Integration, CANBus, AGV/ROV/UAV Integration, ROS (basic)

Standards: MIL-STD-810H, MIL-STD-461G, MIL-STD-1472H

Project Management: MS Project, Jira, Notion, Scrum, Agile/Hybrid Delivery, Stakeholder Management, Risk Management, KPI Tracking, BOM Management, Budget Control

Professional Certifications

- **SOLIDWORKS Professional Certifications** — November 2026
  Sheet Metal, Weldments, Mold Making, Simulation, and API; Dassault Systèmes
- **SOLIDWORKS CAD Design Professional (CSWP)** — December 2024
  Dassault Systèmes
- **Professional Scrum Master I (PSM I)** — July 2026
  Scrum.org
- **Lean Project Management Yellow Belt (LeanPM®-YB)** — August 2025
  Lean Project Management Foundation
- **Lean Six Sigma White Belt (CSSC-LSSWB)** — August 2025
  The Council for Six Sigma Certification
- **IBM Project Management Specialization** — June 2024
  IBM, Coursera
- **Autodesk CAD/CAM/CAE for Mechanical Engineering** — June 2024
  Autodesk, Coursera
- **Collaborative Robotics in Industry Specialization** — June 2024
  L&T EduTech, Coursera
- **Google Project Management Professional Certificate** — February 2024
  Google, Coursera

Achievements

- **1st Runner-Up** — August 2022
  Innovate, IEOM Society AUST Student Chapter
- **Winner** — June 2022
  Kambaii Bangabandhu Robotic Toolkit Design Competition, CRID Robotics Labs, USA
- **Top-5 Industrial Projects Award** — December 2021
  Mujib-100 Industrial Exhibition, IC4IR
- **1st Runner-Up** — March 2021
  TechnoCAD, Innoventure 21, AUST Innovation & Design Club

Leadership & Extracurricular

- **Vice President** — Nov 2022 – May 2023
  AUST Innovation and Design Club
- **Mechanical Sub-Section Member & Design Lead** — Sep 2021 – Sep 2022
  AUST Mars Rover Team
- **Hardware & Design Team Member** — Jan 2022 – Sep 2022
  AUST Rocketry
- **Organizer & Head of Sponsor Team** — Mar 2022 – Jul 2022
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References

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